MERDEKA BELAJAR KAMPUS MERDEKA
OPPO INDONESIA INTERNSHIP
SUMMARY REPORT OF COURSE CONVERSION

ALBERT CRISTIANTO HALIM

22/492268/PA/21099

COMPUTER SCIENCE STUDY PROGRAM

DEPARTMENT OF COMPUTER SCIENCE AND ELECTRONICS

FACULTY OF MATHEMATICS AND NATURAL SCIENCES

UNIVERSITAS GADJAH MADA

2024

CHAPTER I
INTRODUCTION

1.1. Company Profile

OPPO Manufacturing Indonesia is a leading manufacturing facility dedicated to the production and assembly of OPPO's cutting-edge mobile devices. Located in Tangerang, Indonesia, this facility plays a crucial role in ensuring the highest standards of quality and innovation in the smartphone industry. As part of OPPO's global supply chain, the Tangerang-based manufacturing unit is equipped with state-of-the-art production lines, quality control measures, and research-driven processes to maintain its reputation for excellence. OPPO Manufacturing Indonesia is committed to sustainable production practices that minimize environmental impact through energy-efficient processes, responsible sourcing of materials, and waste reduction initiatives. The facility also plays a significant role in job creation and skill development within the local community by providing training programs and career development opportunities. By prioritizing quality, sustainability, and innovation, OPPO Manufacturing Indonesia continues to drive advancements in mobile technology and strengthen its position as a leader in the Indonesian and global smartphone markets.

1.2. MBKM Activity

During the MBKM Independen program, I had the opportunity to work as a Software Engineer Intern at OPPO Indonesia, contributing to two significant projects aimed at improving internal efficiency and digital transformation within the company.

The first project, 360°, is an internal platform designed to facilitate peer performance evaluations among employees. This system enables transparent and structured feedback collection, fostering a stronger feedback culture and improving the overall employee development process. My responsibilities in this project included designing an intuitive user interface (UI) and implementing interactive user experience (UX) elements, ensuring ease of use for employees across different levels of the organization. Additionally, I was involved in backend development, where I worked on setting up the data processing system, performance evaluation metrics, and secure authentication mechanisms. To enhance reliability and accuracy, I developed and conducted unit and module testing to ensure the system's functionality met company requirements. The 360° platform is instrumental in streamlining performance reviews, enhancing HR decision-making, and promoting professional growth within OPPO Indonesia.

The second project, PQPR System Software, was developed to optimize factory changeovers between different products. This system ensures that product transitions on the production line are executed with the best and most efficient configurations, minimizing downtime and improving productivity. It also serves as a centralized database that stores comprehensive product information and employee skill development records over time. My role involved developing an optimized algorithm to generate efficient production line arrangements, integrating a database system to store and track evolving operator skill levels, and ensuring real-time updates to reflect changes in factory processes. Additionally, I collaborated with factory engineers, system analysts, and IT teams to fine-tune the solution for seamless integration into existing workflows. The system plays a crucial role in enhancing production efficiency, reducing operational bottlenecks, and ensuring that workforce capabilities are utilized effectively.

These projects were directly aligned with key academic disciplines, enabling me to apply theoretical knowledge to real-world industry scenarios. The end-to-end development of both projects corresponded to **Manajemen Proyek Teknologi Informasi (MII21-3506),** where I was responsible for project planning, task management, and timeline coordination. My work in backend development, including API integration, database structuring, and request optimization, was related to Internship: **Pengembangan Backend (MII21-3014)**. To maintain system integrity and functionality, I performed rigorous unit testing, debugging, and validation, aligning with Internship: **Pengujian Unit dan Modul Proyek (MII21-3015)**. Furthermore, working within a corporate environment required effective communication and collaboration with cross-functional teams, addressing challenges and presenting solutions, which matched the objectives of **Softskill: Kemampuan Berkomunikasi (MII21-4010)** and **Softskill: Kemampuan Bekerjasama dan Kolaborasi (MII21-4012)**.

This internship experience significantly contributed to OPPO Indonesia’s digital transformation efforts while providing me with hands-on experience in software engineering, system optimization, and corporate collaboration. The opportunity to develop real-world applications, integrate solutions into existing business processes, and work alongside industry professionals has strengthened my technical and interpersonal skills, preparing me for future challenges in the field of technology and innovation.

CHAPTER II
TRANSCRIPT AND COURSES

2.1. Transcript

UNIVERSITAS GADJAH MADA
FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM

TRANSKRIP SEMENTARA
Semester Genap 2024/2025

Nama : ALBERT CRISTIANTO HALIM
NIM : 22/492268/PA/21099
Prodi : S1 ILMU KOMPUTER
DPA : Yunita Sari, S.Kom., M.Sc., Ph.D.

No	Kode	Mata Kuliah	SKS	Kelompok	Jenis	Ke	Nilai
1	MFF1011	**Fisika Dasar 1** Kelas: IUP2	3.00	MKK	Wajib	1	A
2	MMM1101	**Kalkulus 1** Kelas: IUP2	3.00	MKK	Wajib	1	C
3	MKK1101	**Kimia Dasar 1** Kelas: IUP-1	3.00	MKK	Wajib	1	A/B
4	MII211201	**Pemrograman** Kelas: IUP2	3.00	MKK	Wajib	1	A
5	MII211202	**Praktikum Pemrograman** Kelas: CSA	1.00	MKK	Wajib	1	B+
6	MII211001	**Aljabar Linier Fundamental** Kelas: CSA	2.00	MKK	Wajib	1	A
7	MII211003	**Bahasa Indonesia dan Etika Ilmiah** Kelas: CSA	2.00	MKK	Wajib	1	A/B
8	MII211002	**Logika Informatika** Kelas: CSA	2.00	MKK	Wajib	1	A
9	UNU1001	**Agama Katholik** Kelas: IUP	2.00	MPK	Wajib	1	A-
10	MII211203	**Algoritma dan Struktur Data** Kelas: CSA	3.00	MKB	Wajib	1	A-
11	MII211005	**Integral dan Persamaan Diferensial** Kelas: CSA	3.00	MKK	Wajib	1	A-
12	UNU3000	**Kewarganegaraan** Kelas: CS	2.00	MPK	Wajib	1	A
13	MII211006	**Matematika Diskrit** Kelas: CSA	3.00	MKK	Wajib	1	A/B
14	MII211601	**Organisasi dan Arsitektur Komputer** Kelas: CSA	2.00	MKB	Wajib	1	A-
15	UNU1010	**Pancasila** Kelas: CS	2.00	MPK	Wajib	1	A
16	MII211007	**Pengantar Statistika** Kelas: CSA	2.00	MKB	Wajib	1	A
17	MII212003	**Pengembangan Startup Digital** Kelas: CS	2.00	MKK	Wajib	1	A
18	MII211204	**Praktikum Algoritma dan Struktur Data** Kelas: CSA	1.00	MKB	Wajib	1	A/B
19	MII211602	**Sistem Digital** Kelas: CSA	2.00	MKK	Wajib	1	A
20	MII211004	**Bahasa Inggris** Kelas: CS	2.00	MBB	Wajib	1	A

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No	Kode	Mata Kuliah	SKS	Kelompok	Jenis	Ke	Nilai
21	MII212201	**Analisis Algoritma dan Kompleksitas** Kelas: CSA	3.00	MKK	Wajib	1	A
22	MII212501	**Basis Data** Kelas: CSA	3.00	MKK	Wajib	1	A-
23	MII212502	**Praktikum Basis Data** Kelas: CSA	1.00	MKK	Wajib	1	A
24	MII212606	**Internet of Things dan Aplikasinya** Kelas: CS	3.00	MKB	Pilihan	1	B
25	MII212601	**Jaringan Komputer** Kelas: CSA	2.00	MKK	Wajib	1	A-
26	MII212401	**Kecerdasan Artifisial** Kelas: CSA	3.00	MKK	Wajib	1	A/B
27	MII212603	**Praktikum Sistem Komputer dan Jaringan** Kelas: CSA	1.00	MKB	Wajib	1	A
28	MII212001	**Probabilitas dan Proses Stokastik** Kelas: CSA	2.00	MKK	Wajib	1	A
29	MII212602	**Sistem Operasi** Kelas: CSA	2.00	MKK	Wajib	1	A
30	MII213208	**Computational Thinking** Kelas: CS	2.00	MKB	Pilihan	1	A
31	MII212202	**Bahasa dan Otomata** Kelas: CSA	3.00	MKK	Wajib	1	A-
32	MII212610	**Komputasi Awan** Kelas: CS	3.00	MKB	Pilihan	1	A/B
33	MII212209	**Kriptografi dan Keamanan Informasi** Kelas: CSA	3.00	MKK	Wajib	1	A
34	MII212203	**Metode Numerik** Kelas: CSA	2.00	MKK	Wajib	1	A/B
35	MII212503	**Metode Rekayasa Perangkat Lunak** Kelas: CSA	2.00	MKK	Wajib	1	A-
36	MII212402	**Pembelajaran Mesin** Kelas: CSA	3.00	MKK	Wajib	1	A
37	MII212506	**Pengembangan Perangkat Lunak Scalable** Kelas: CS	3.00	MKB	Pilihan	1	A-
38	MII212206	**Penglihatan Komputer dan Analisis Citra** Kelas: CS	3.00	MKB	Pilihan	1	A-
39	MII212504	**Workshop Implementasi Rancangan Perangkat Lunak** Kelas: CSA	2.00	MKK	Wajib	1	A/B
40	MII213011	**Internship: Pengembangan Fitur dan Modul Proyek** Kelas: IMBKM	4.00	MKB	Pilihan	1	A

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No	Kode	Mata Kuliah	SKS	Kelompok	Jenis	Ke	Nilai
41	MII213016	**Internship: Pengujian Integrasi dan Sistem** Kelas: IMBKM	3.00	MKB	Pilihan	1	A
42	IEE3593	**Introduction to Big Data Analysis** Kelas: EXC	3.00		Pilihan	1	A
43	MII213002	**Metodologi Penelitian Ilmu Komputer** Kelas: CSA	2.00	MKK	Wajib	1	A
44	MII213401	**Pembelajaran Mesin Mendalam** Kelas: CSA	3.00	MKK	Wajib	1	A
45	MII213501	**Proyek Rekayasa Perangkat Lunak** Kelas: CSA	3.00	MKK	Wajib	1	A-
46	MII214023	**Global Employability skill** Kelas: IMBKM	2.00		Pilihan	1	A
47	MII213001	**Kelas Seminar** Kelas: SSan	1.00	MKK	Wajib	1	A
48	MII212505	**Pengembangan UIUX Frontend** Kelas: IMBKM	3.00	MKB	Pilihan	1	A
49	MII213014	**Internship: Pengembangan Back-end** Kelas: IMBKM	4.00	MKB	Pilihan	1	
50	MII213015	**Internship: Pengujian Unit dan Modul Proyek** Kelas: IMBKM	3.00	MKB	Pilihan	1	
51	MII214010	**Softskill: Kemampuan Berkomunikasi** Kelas: IMBKM	2.00	MKB	Pilihan	1	
52	MII214012	**Softskill: Kemampuan Bekerjasama dan Kolaborasi** Kelas: IMBKM	2.00	MKB	Pilihan	1	
53	UNU242033	**Literasi Kesehatan** Kelas: KOMCS	2.00		Wajib	1	
54	MII214001	**Proposal Skripsi** Kelas: TACS	2.00	MPB	Wajib	1	
55	MII214002	**Skripsi** Kelas: TACS	6.00	MPB	Wajib	1	
		Jumlah SKS	136				

Indeks Prestasi Sementara (IPS) : / =
Indeks Prestasi Kumulatif (IPK) : 433.5/115 = 3.77

Yogyakarta, 26 Maret 2025
Dosen Pembimbing Akademik

Yunita Sari, S.Kom., M.Sc., Ph.D.
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[FO-Transkrip Sementara]

2.2. Course List

No	Code	Course	SKS
1	MII21-3506	Manajemen Proyek Teknologi Informasi	3
2	MII21-3014	Internship: Pengembangan Backend	4
3	MII21-3015	Internship: Pengujian Unit dan Modul Proyek	3
4	MII21-4010	Softskill: Kemampuan Berkomunikasi	2
5	MII21-4012	Softskill: Kemampuan Bekerjasama dan Kolaborasi	2
Total MBKM			14

Table 2.2.1. Course Converted

CHAPTER III

MBKM - Hardskill: Pengembangan Backend

Name	: Albert Cristianto Halim
NIM	: 22/492268/PA/21099
Academic Supervisor	: Yunita Sari, S.Kom., M.Sc., Ph.D
MBKM Program Supervisor	: Dr. Azhari, M.T
Course	: Pengembangan Backend
Course Code	: MII21-3014
Total Credits	: 4 SKS
Internship Title	: MBKM Mandiri OPPO Indonesia
Internship Duration	: January 2025 – April 2025

3.1.Syllabus

Students will be able to design and implement a system, subsystem, application, or prototype related to backend/server development based on specifications from their internship/MBKM activities. The description of backend/server development includes the following aspects:

1. **General Description** – Explanation of the prototype, application, system, or product on the backend side.

2. **Development Tools & Technologies** – Details on the tools, libraries, add-ins, plugins, or hardware used.

3. **System/Server Configuration** – Explanation of server/system setup, sample data configuration, dataset usage, or database structure.

4. **Source Code & Implementation** – Inclusion of source code (SQL, NoSQL), output display, and explanations of functionalities, specifications, and system design.

3.2. Activities

3.2.1. 360 Peer-to-Peer Grading Project

The **360 Peer-to-Peer Grading System** is a web application designed to support comprehensive peer and self-evaluation processes in educational or organizational contexts. The system enables both **admins** and **users** to participate in a structured grading process within defined time periods.

Key functionalities of the system include:

- **Admin capabilities**:
  - Creation and management of user accounts.
  - Creation and scheduling of grading periods.
  - Access to result recaps in various formats (PDF, XLSX).
- **User capabilities**:
  - Access to assigned peer-to-peer and self-grading tasks.
  - Ability to export personal grading results in PDF format.

The backend system is responsible for user management, grading data handling, secure authentication, and generation of recap documents.

a. **System Configuration**

- **Server Setup:**
  - Backend services run on a Node.js environment.
  - The database is a PostgreSQL instance, either locally or hosted on Supabase/PostgreSQL server.
  - Authentication and session management are handled securely via NextAuth.js.
- **Database Structure (Entity Relationship Design)**

The database design of the **360 Peer-to-Peer Grading System** is structured to efficiently support the management of users, grading periods, peer/self-assessments, and their associated data relationships.

The design is implemented on a **PostgreSQL** relational database, ensuring data integrity, consistency, and scalability. There are three primary entities:

1. **Users**

   The **User** entity stores all information about participants in the system. This includes both **administrators** and **standard users** (participants who perform peer and self-assessments).

Column	Description
`id`	Unique identifier for each user (PK).
`name`	Name of the user.
`nik`	A unique identifier
`bod`	Birth date of the user.
`password`	Encrypted password for authentication.
`level`	User's organisational level.
`title`	Job title or position of the user.
`department`	Department or division where the user belongs.
`is_admin`	Boolean flag to distinguish admin users (true) from regular users (false).
`is_active`	Boolean flag indicating whether the user account is active.

2. **AssessmentPeriod**

   The **AssessmentPeriod** entity defines the periods during which peer-to-peer and self-assessments are conducted.

   A period represents a **specific timeframe** where participants complete evaluations.

Column	Description

id	Unique identifier for each assessment period (PK).
created_at	Timestamp when the period record was created.
period	Descriptive name of the period (e.g., “Mid-Year 2025”).
year	The year in which the assessment period takes place.
is_active	Boolean flag indicating whether the period is currently active.
start_date	Date and time when the assessment period starts.
end_date	Date and time when the assessment period ends.

3. **AssesmentSingular**

The **AssessmentSingular** entity stores individual assessment records i.e., every instance where a user assesses another user (or themselves).

It is the **core table** where **grading data** is captured, supporting both **peer-to-peer** and **self-assessment** processes.

Column	Description
id	Unique identifier for each assessment period (PK).
id_user	The user being assessed (Foreign Key → User.id).
id_assessor	The user conducting the assessment (Foreign Key → User.id).
id_assessment_period	The period in which this assessment took place (Foreign Key → AssessmentPeriod.id).
sub_date	Timestamp when the assessment was submitted.
kritik	Written critique or feedback.
saran	Written suggestions or recommendations.
q_1 until q_27	Numeric scores for 27 predefined assessment questions (double precision).
is_sub	Boolean flag indicating whether the assessment has been submitted (true) or is still in progress (false).

`relationship`	Describes the relationship between assessor and assessee (e.g., peer, subordinate, supervisor).

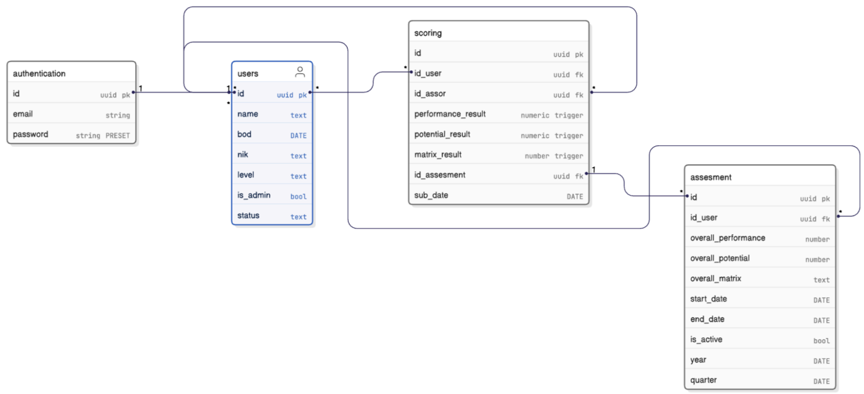

Figure 3.2.1 ERD Diagram

b. Source Code & Implementation (Functional Overview)

The backend system implements the following main functionalities:

- **User Management (Admin)**
  Admins can create, update, and manage user accounts, assigning them as either administrators or standard users.

- **360 Period Management (Admin)**
  Admins define grading periods by setting a name, start date, and end date. These periods determine when users can perform grading activities.

- **Peer Assignment (Admin)**
  Admins assign users to grade their peers and themselves during a given period. This mapping ensures every user has both self and peer evaluations to complete.

- **Grading Submission (User)**
  Users access their assigned grading tasks and submit scores and comments based on predefined criteria.

- **Document Generation and Export**
  Both admins and users can export grading recaps:
  - **Admins**: Download recap summaries in PDF or XLSX formats (covering entire periods or individual users).
  - **Users**: Download personal grading results in PDF format.
- **Authentication & Authorization**
  The backend uses NextAuth.js for secure login, session management, and role-based access control.

3.2.2. PQPR Software Project

The PQPR Software is a web-based application designed to improve process quality analysis and layout standardization in assembly manufacturing environments. The system automates the analysis of process and layout differences across multiple assembly models by leveraging Process Quality and Planning Route (PQPR) metrics. Through this system, manufacturing operators and engineers can compare process steps, machine layouts, cycle times, and operator skills between different models, helping to ensure standardized practices across production lines.

The PQPR Software successfully improved process consistency from **63% to 80%** and layout standardization from **64% to 80%** by utilizing enhanced visual analytics and automated model comparison tools.

The system consists of:

- **Model Management**: creation and maintenance of assembly models, their manpower, market, and type categories.
- **Machine & Work Element Registry**: records of all machines and process elements used in assembly lines.

- **Cycle Time and SOP (Standard Operating Procedure)**: management of production cycle times and standard processes associated with different models.

- **Operator Management**: management of operator details and their associated skills.

- **Comparison Dashboard**: tool to compare process routes and layouts between different models.

a. System Configuration

- **Server Setup:**
  - **Frontend** (Next.js): Deployed via Node.js environment on Docker containers.
  - **Backend** (FastAPI): Deployed on a separate Docker container, connected to the PostgreSQL database through a secure internal network.
  - **Database:** PostgreSQL configured with role-based access control (RBAC) and connection pooling.
  - **Sample Data:** Predefined models, machine registries, SOPs, and work elements were seeded into the database for testing and demonstration purposes.
  - **Version Control:** GitHub was used for source code versioning and team collaboration.

b. Database Structure (Entity Relationship Design)

The database schema for the PQPR Software captures the relationships between models, machines, processes, operators, and standard procedures essential for analyzing process quality and layout standardization. Below are the key tables and their attributes.

1. **Model**

The **Model** entity stores key details about each assembly model, including its production capacity, manpower, and market information. It serves as a core reference for comparing process times and layouts between models.

Column	Description
`id`	Unique identifier for the model
`model_name`	Name of the model
`market_name`	Market segment of the model
`total_manpower`	Total manpower required
`type_category`	Type classification of the model
`status_family`	Status group/family
`uph`	Units per hour (production rate)

2. **Machines**

The **Machines** entity captures machine details in the assembly line, including unique codes, names, areas, and process sequences. This allows modeling of physical layouts and process flows.

Column	Description
`id`	Unique identifier for the machine
`code`	Machine code
`process_sequence`	Process sequence number
`machine_name`	Machine name
`area`	Factory area where machine is placed (Installation/ Testing)

3. **ModelPRLayout**

The **ModelRPLayout** entity defines the spatial layout of machines for a specific model. It maps machine-to-machine relationships and positions, enabling layout comparisons between models.

Column	Description
`id`	Unique identifier for layout
`id_machine`	References Machines.id
`id_model`	References Model.id
`side_a_machine`	References Machines.code (side A machine)
`side_b_machine`	References Machines.code (side B machine)
`side_a_position`	Position of side A machine
`side_b_position`	Position of side B machine

4. **WorkElement**

The **WorkElement** entity captures the smallest unit of work within the assembly process. It defines each task's sequence, name, and area, providing a building block for SOPs and cycle time tracking.

Column	Description
`id`	Unique identifier of Work Element
`code`	Work element code
`process_sequence`	Sequence in the process
`work_element_name`	Name of the work element
`area`	Work area (Installation/ Testing)

5. **CycleTime**

The CycleTime entity records the time required to complete each work element in a specific model. It is used to analyze process efficiency and compare timing across models.

Column	Description
`id`	Unique identifier of work element cycle time
`id_model`	References Model.id
`id_work_element`	References WorkElement.id
`model_name`	Name of model
`code`	References WorkElement.code
`work_element`	References WorkElement.work_element_name
`area`	References WorkElement.area
`cycle_time`	Cycle time (seconds or minutes)

6. **SOP**

The **SOP** entity defines the formalized instructions for performing work elements in the assembly process. It links SOPs directly to specific models and supports operator training and process standardization.

Column	Description
`id`	Unique identifier of SOP
`no_sop`	SOP number
`name`	SOP name
`id_model`	References Model.id
`model`	SOP associated model name

7. **SOPWE**

The **SOPWE** entity links standard operating procedures to the corresponding work elements that they cover, enabling a many-to-many relationship between SOPs and tasks.

Column	Description

id_sop	References SOP.id
id_work_element	References WorkElement.id

8. Operator

The **Operator** entity stores information about assembly line workers, including their identification, names, and current status. It is essential for tracking operator involvement in SOPs.

Column	Description
id	Unique identifier of Operator
nik	Operator NIK (Employee ID)
name	Operator name
status	Operational status

9. OperatorSkill (Conjuction Table)

The **OperatorSkills** entity maps the skills of operators to the SOPs they are certified to execute, supporting skill matrix tracking and competency management.

Column	Description
id_operator	References Operator.id
id_sop	References SOP.id

10. Comparison

The **Comparison** entity stores records of model-to-model comparisons, which allow users to analyze differences in process quality and layout between two distinct models.

Column	Description

`id`	Unique identifier
`first_type`	First model being compared
`second_type`	Second model being compared

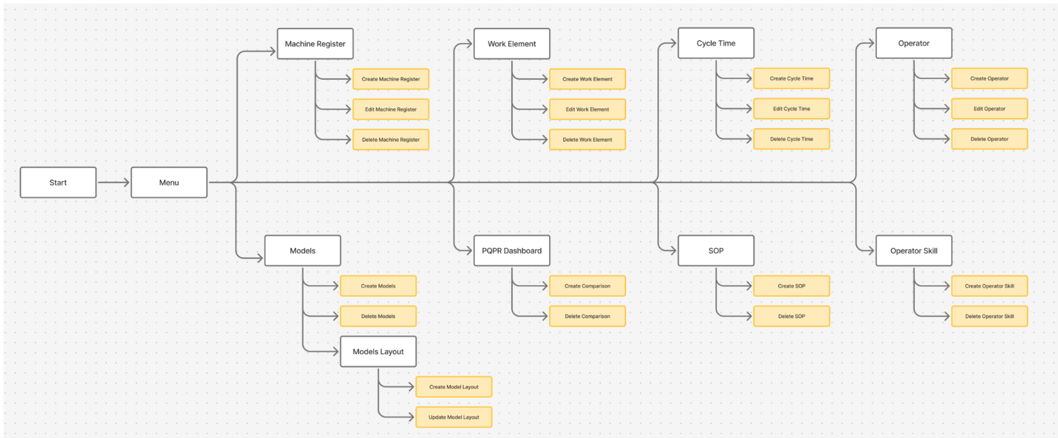

Figure 3.2.2 PQPR Application Userflow

3.3.Fundamental Concepts Related to The Syllabus

The activities carried out during the internship at OPPO Indonesia are closely aligned with the core concepts outlined in the syllabus of the *Pengembangan Backend* course. These fundamental concepts form the backbone of modern backend development and were consistently applied throughout the projects:

1. **System Design and Architecture**
   The projects emphasized the ability to design scalable and maintainable systems. The 360 Peer-to-Peer Grading System and PQPR Software both required robust architectural planning, covering database schema design, API structuring, and service modularization. Entity-Relationship Diagrams (ERD) were extensively utilized to ensure clarity in data modeling and relationships.

2. **Database Management and Optimization**
   Relational database management concepts were applied through PostgreSQL, including normalization, indexing, and role-based access control (RBAC). Features such as Row Level Security (RLS) and connection pooling ensured data integrity, optimized queries, and secure access management — directly corresponding to syllabus components like database configuration and dataset

usage.

3. **Backend Development Tools and Technologies**
   Modern frameworks and technologies such as FastAPI, Node.js, NextAuth.js, and Supabase were deployed to streamline backend development. These tools supported rapid development cycles, secure authentication mechanisms, and seamless API integration — directly reflecting the syllabus requirement to apply development tools and libraries in practice.

4. **Source Code Implementation and Functional Integration**
   A clear understanding of backend functionalities was demonstrated through user management systems, document generation modules, and grading workflows. These implementations required knowledge of RESTful API development, secure session management, and server-client communication aligning with the syllabus objective to produce functional prototypes and applications.

5. **Security and Access Control**
   Security principles such as encrypted authentication, session management, and granular role validation were implemented using NextAuth.js and database-level policies. These measures ensured data protection, a crucial aspect of backend development emphasized in the syllabus.

3.4.Result and Discussion

The implementation of features and modules during the internship for Internship: Pengembangan Backend (MII21-3014) yielded significant results, demonstrating the practical application of system design and development concepts. The projects achieved its objective of facilitating a secure, efficient, and paperless documentation system for OPPO Indonesia.

3.4.1. Database Architecture and Role Management

The implementation of a well-designed database architecture with a robust Entity-Relationship Diagram (ERD) ensured a strong foundation for managing user roles, document data, and activity logs. The integration of Row Level Security (RLS) at the database level successfully restricted data access based on user roles, ensuring data integrity and security. The role management system effectively distinguished between Admins, Users. allowing for precise control over functionalities like document creation, user management, and document authorization.

3.4.2. Request and Load Optimization

Client-Side Rendering (CSR) techniques were employed to optimize the performance of the application. Actions like loading user lists, creating documents, and navigating dashboards were streamlined through dynamic data fetching and on-demand rendering. These optimizations reduced server load, improved responsiveness, and ensured a smooth user experience.

3.4.3. PDF Generation and Document Management

The development of a PDF Compiler enabled the application to generate structured and professional PDF documents for Peer-to-peer evaluation result. The integration of digital duty stamps and signature authorization ensured compliance with administrative standards, while real-time previews allowed users to validate inputs before finalizing documents.

3.4.4. Security Measures

The implementation of multi-layered security measures, including front-end role validation and database-level RLS, safeguarded sensitive data and functionalities. Unauthorized access was prevented through strict role-based controls, ensuring that only authorized users could perform actions like document approval or user management.

3.5. Conclusion

The independent internship program (MBKM Mandiri) at OPPO Indonesia provided an invaluable opportunity to apply backend development theories to practical, industry-relevant projects. Through the implementation of the 360 Peer-to-Peer Grading System and the PQPR Software Project, key competencies such as system design, database management, secure authentication, and process optimization were successfully exercised.

The experience fostered the development of robust technical skills, including proficiency in backend frameworks (Node.js, FastAPI), database systems (PostgreSQL, Supabase), and modern development workflows (Git). In addition, the projects emphasized critical soft skills such as collaborative problem-solving, project management, and technical communication within a professional setting.

Overall, the internship not only reinforced the academic concepts outlined in the *Pengembangan Backend* syllabus but also bridged the gap between classroom learning and industry application. The outcomes achieved during the internship — from secure document management systems to standardized manufacturing process tools — reflect a holistic growth in backend engineering capabilities, preparing the participant for future contributions in the field of information technology and system development.

CHAPTER IV

MBKM - Hardskill: Manajemen Proyek Teknologi Informasi

Name	: Albert Cristianto Halim
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Academic Supervisor	: Yunita Sari, S.Kom., M.Sc., Ph.D
MBKM Program Supervisor	: Dr. Azhari, M.T
Course	: Manajemen Proyek Teknologi Informasi
Course Code	: MII21-3506
Total Credits	: 3 SKS
Internship Title	: MBKM Mandiri OPPO Indonesia
Internship Duration	: January 2025 – April 2025

4.1.Syllabus

1. Project & IT organization
2. Project proposal & procurement
3. Project tools & scheduling algorithms
4. Cash, budget estimation, & monitoring
5. Human resources allocation & promotion
6. project Monitoring
7. Project portfolio & risk management
8. Project certification.

4.2. Activities

Throughout the internship, project management activities were carried out using **GitHub Projects**, an integrated project management tool that facilitated collaborative planning, task tracking, and progress monitoring. GitHub Projects was utilized for both the **360 Peer-to-Peer Grading System** and **PQPR Software Project** to ensure systematic development and effective coordination.

The key project management activities included:

1. **Project Planning and Task Breakdown**

   Each project was initiated with a structured breakdown of tasks using GitHub Project boards. Epics, milestones, and features were defined and organized into **Kanban-style columns**: *To Do*, *In Progress*, *Review*, and *Done*. Tasks were represented as GitHub issues, each tagged with labels such as *backend*, *frontend*, *database*, and *testing* to ensure clarity in scope and responsibility.

2. **Team Collaboration and Assignment**

   Issues and tasks were assigned to specific team members, ensuring clear ownership and accountability. GitHub's assignee and mention features streamlined communication between developers, while comments and issue threads served as centralized discussion spaces for clarifying requirements and documenting decisions.

3. **Progress Monitoring and Reporting**

   GitHub Projects enabled real-time progress monitoring through visual boards and automated workflows. Task completion status, blockers, and sprint progress were continuously updated, allowing stakeholders to track project health efficiently. Weekly updates and sprint retrospectives were documented using GitHub Issues and Pull Requests, ensuring transparency in deliverables.

4. **Integration with Version Control**

   GitHub Projects was directly linked to the source code repositories, allowing for seamless integration of project management with development workflows. Pull requests were associated with issues, enabling automatic closure of tasks upon successful code merges. This integration enhanced traceability between code changes and project milestones.

4.3.Fundamental Concepts Related to The Syllabus

The project management activities performed during the internship directly applied and reflected the key concepts outlined in the *Manajemen Proyek Teknologi Informasi* syllabus. The following fundamental concepts were integrated throughout the project lifecycle of the 360 Peer-to-Peer Grading System and PQPR Software Project:

1. **Project and IT Organization**

   The projects were structured following a clear organizational hierarchy of roles and responsibilities. Team members were assigned specific tasks via GitHub Issues, ensuring distinct role definitions and smooth collaboration. The systematic organization of tasks and milestones reflected structured project governance in IT environments.

2. **Project Proposal and Procurement**

   Initial project scoping, requirement gathering, and task planning were conducted collaboratively using GitHub Projects. Each proposed feature or module was documented as an issue or epic, functioning as a lightweight project proposal. Resource procurement — including datasets, development environments (Docker, PostgreSQL), and libraries (Next.js, FastAPI) — was planned and executed in alignment with project objectives.
3. **Project Monitoring**

   Continuous project monitoring was facilitated through GitHub's visual project boards, issue tracking, and automated progress updates. Regular sprint reviews and weekly stand-up summaries were documented via GitHub Issues and Pull Requests, ensuring systematic progress assessment and risk identification.
4. **Project Certification (Project Closure)**

   Upon completion of major milestones, deliverables were documented, merged, and closed within GitHub. This formal project closure process ensured that both code and documentation were finalized, reviewed, and archived systematically, embodying principles of project certification and handover.

4.4.Result and Discussion

The adoption of GitHub Projects for managing the internship projects yielded several significant outcomes:

1. **Improved Task Organization and Clarity**
   The breakdown of project components into granular, well-defined tasks ensured a clear understanding of deliverables and timelines. Both the 360 Grading System and PQPR Software Project benefited from organized workflows that minimized ambiguity and streamlined task allocation.

2. **Enhanced Team Collaboration and Communication**
   Centralized communication via GitHub Issues reduced the reliance on external messaging platforms, ensuring that all discussions, decisions, and progress updates were documented and easily accessible. This practice facilitated better alignment among team members, especially during remote collaboration.

3. **Increased Transparency and Accountability**
   The visual representation of task status on Kanban boards improved project visibility for all stakeholders. This transparency promoted accountability, as task ownership and completion statuses were clearly displayed and traceable.

4. **Efficient Integration with Development Workflow**
   The direct link between issues, pull requests, and commits streamlined the transition from planning to implementation. Developers could easily associate code changes with corresponding tasks, reducing context-switching and improving productivity.

5. **Timely Delivery and Continuous Monitoring**
   By leveraging milestones and automated project workflows, deadlines were met effectively, and sprint progress was continuously monitored. Both projects achieved their targeted deliverables within the internship timeframe (January–April 2025), demonstrating effective project execution.

4.5.Conclusion

The project management activities conducted during the internship at OPPO Indonesia, particularly through the utilization of **GitHub Projects**, provided a practical and comprehensive application of the concepts outlined in the *Manajemen Proyek Teknologi Informasi* syllabus. From task planning and resource allocation to progress monitoring and project documentation, all phases of project management were effectively executed.

The integration of GitHub Projects into the software development lifecycle ensured clear task visibility, streamlined communication, and efficient tracking of deliverables. These practices facilitated the successful completion of both the 360 Peer-to-Peer Grading System and PQPR Software Project within the allocated timeframe and scope.

The experience not only strengthened technical project management skills but also demonstrated how modern agile-based tools can optimize collaboration and ensure project success in real-world software development environments. These learnings and practices will serve as a strong foundation for future professional endeavors in information technology project management.

CHAPTER V

MBKM - Hardskill: Pengujian Unit dan Modul Proyek

Name	: Albert Cristianto Halim
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Academic Supervisor	: Yunita Sari, S.Kom., M.Sc., Ph.D
MBKM Program Supervisor	: Dr. Azhari, M.T
Course	: Pengujian Unit dan Modul Proyek
Course Code	: MII21-3015
Total Credits	: 3 SKS
Internship Title	: MBKM Mandiri OPPO Indonesia
Internship Duration	: January 2025 – April 2025

5.1. Syllabus

Students will be able to design unit testing for software, modules, subsystems, or prototypes and conduct testing using test data from their internship/MBKM activities. The description of unit/module testing may include the following aspects:

1. **General Testing Description** – Explanation of testing objectives, strategies, and methods used.

2. **Testing Tools & Technologies** – Details on the tools, libraries, add-ins, plugins, or hardware used.

3. **List of Units/Features/Modules Tested** – Identification of the components that will undergo testing.

4. **Test Case Design** – A table or list of test cases, including columns such as: (Test case number, Name of the unit/module/component/subprogram, Test data, Testing method/process, Expected test results

5. **Testing Results & Analysis** – Documentation of test outputs, validation/verification, and discussion of the results.

5.2. Activities

During the internship period, unit and module testing activities were conducted on various core features of the PQPR Software Project. The testing process involved both **API endpoint validation** and **feature functionality testing** using Postman, with a focus on CRUD operations, data retrieval accuracy, and system consistency.

The tested endpoints included:

- **Standard Operating Procedures (SOP)**
  Endpoint: GET /sop
  Verified the retrieval of SOP records, their relationships to work elements (sop_we), and data accuracy such as SOP number, model name, and associated tasks.

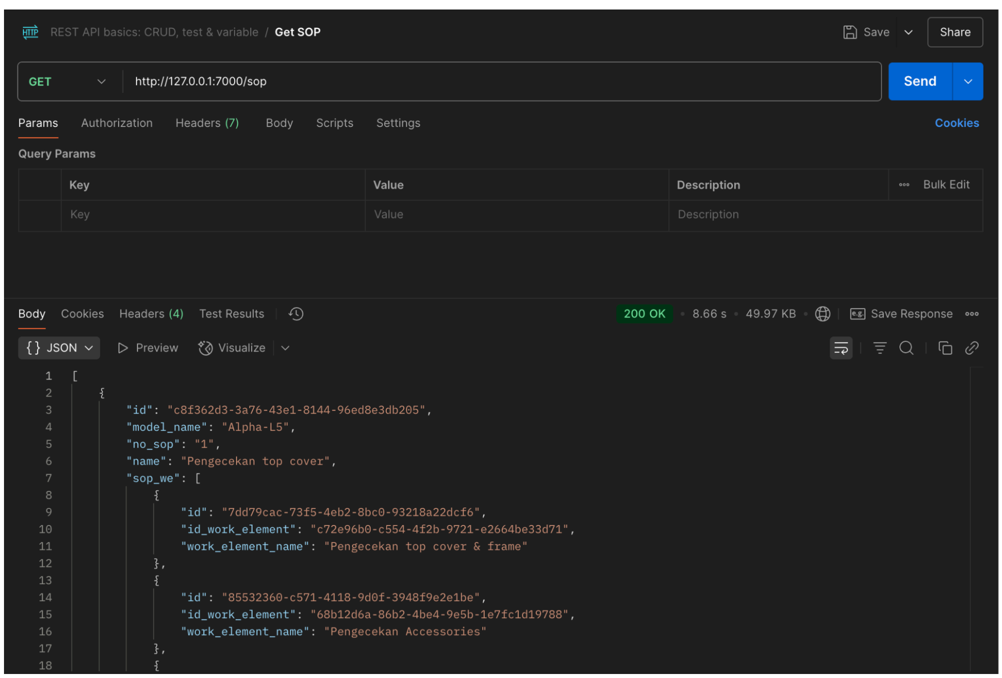

Figure 5.2.1 Postman Request SOP

- **Machines**
  Endpoint: GET /machines

Validated the retrieval of machine data, including machine code, process sequence, machine name, and area (Installation/Testing). Ensured consistency with the factory layout model.

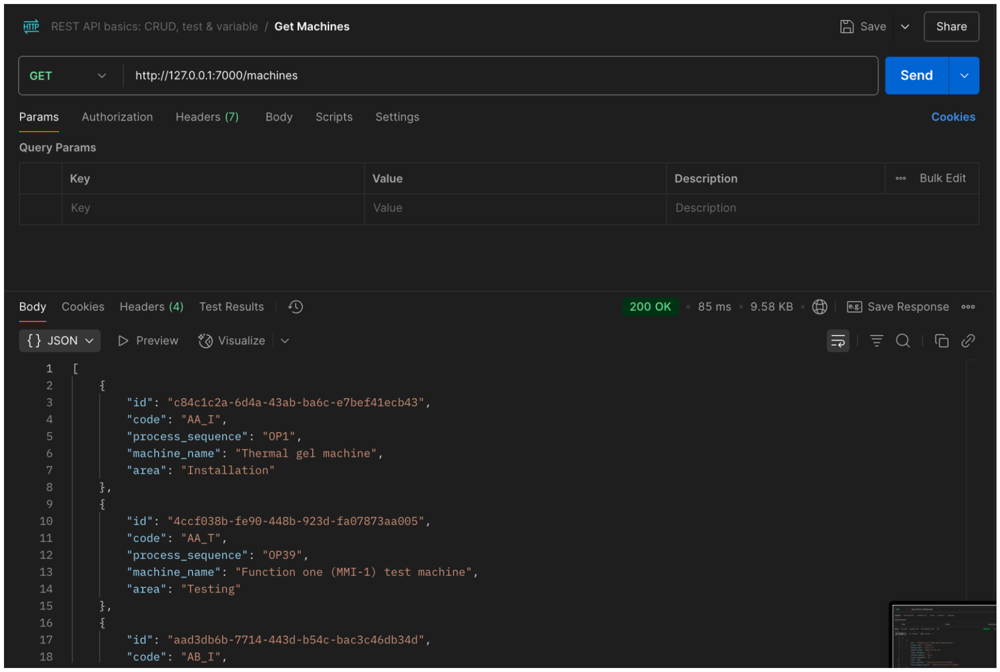

Figure 5.2.1 Postman Request Machines

- **Model**

 Endpoint: GET /model

 Tested model data retrieval, which includes model details (name, market segment, total manpower, production rate — UPH) and last update timestamps. Confirmed that records reflected recent updates accurately.

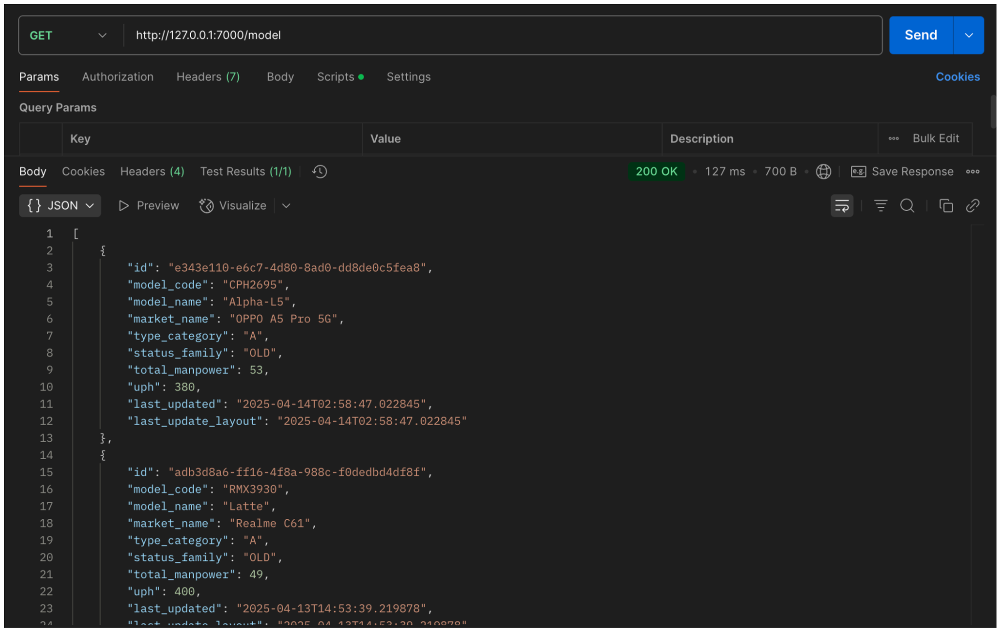

Figure 5.2.1 Postman Request Model

These tests aimed to confirm that the backend API correctly returned well-structured JSON responses, maintained referential integrity between entities (via foreign keys), and aligned with the functional expectations of the PQPR Software's comparison and analysis features.

In addition to API testing, **integration testing** was also performed to validate interactions between database entities, such as the linkage between models, machines, and SOPs, ensuring that the application workflows operated seamlessly.

5.3. Fundamental Concepts Related to The Syllabus

5.3.1. Purpose of Testing

The primary purpose of testing was to validate that each feature and module, as well as their integrations, met the defined functional and business requirements. Testing ensured that the system operated as intended with minimal errors, that components interacted seamlessly, and that data consistency was maintained across all modules. Security mechanisms, such as role-based access control, were also verified to ensure compliance with user role restrictions. This purpose-driven testing approach was critical to identifying potential issues and ensuring alignment with real-world use scenarios.

5.3.2. Testing Strategies and Methods

The testing strategy was divided into two main phases: unit testing and integration testing. Unit testing focused on validating individual features and modules, such as login functionality, user management, document workflows, and PDF generation. Integration testing assessed the interaction between these components to ensure data flowed accurately and workflows were consistent. This modular-to-system-wide testing approach ensured comprehensive coverage, from isolated feature validation to interconnected system performance.

5.3.3. Testing Features and Modules

Testing concentrated on the core features of the PQPR and 360 Peer-to-peer evalutation application. The login system was tested for secure user authentication and session management. The user management module underwent rigorous testing for creating, editing, and deleting user accounts, ensuring data accuracy. The PDF generation module was evaluated for formatting and accuracy, including the integration of digital duty stamps and signature authorization. Lastly, role management was tested to ensure access control adhered to user permissions, such as restricting sensitive features to Admins.

5.3.4. Test Case Design

Test cases were systematically designed to ensure all functionalities and workflows were thoroughly tested. Each test case specified the feature or module being tested, the input data (both valid and invalid), the testing method or approach, and the expected outcomes. By using this structured approach, every feature of the application was validated against predicted outcomes, ensuring that no critical functionality was overlooked during testing.

5.4. Result and Discussion

5.4.1. Accurate Data Retrieval

All API endpoints returned the expected datasets, with complete fields and valid foreign key relationships. For instance, the GET /sop endpoint accurately linked SOPs to their relevant work elements, as confirmed by matching IDs and descriptive task names.

5.4.2. Correct Entity Relationships

The Model, Machines, and SOP entities maintained proper referential integrity. The GET /model responses demonstrated that model records correctly reflected associated market information, manpower data, and updated layout timestamps critical for the PQPR's process comparison workflows.

5.4.3. API Performance

The API endpoints consistently returned HTTP status **200 OK** responses, with acceptable response times (e.g., ~85 ms for Machines, ~127 ms for Models), ensuring responsive interactions. The JSON response sizes (e.g., ~50 KB for SOPs) were well-optimized for efficient frontend rendering.

5.4.4. Validation of Complex Modules

Through structured API testing, more complex modules such as SOP-to-WorkElement mappings and Model-to-Machine layout definitions were validated to ensure consistency in hierarchical data representation. These validations confirmed the system's readiness for real-world manufacturing process analyses.

5.5. Conclusion

The unit and module testing activities conducted during the internship at OPPO Indonesia successfully validated the core functionalities and API endpoints of the PQPR Software Project. Through comprehensive testing including API validation, data integrity verification, and performance assessment the project demonstrated compliance with functional requirements and ensured system reliability.

The testing outcomes confirmed that features such as SOP management, machine registry, and model configuration operated accurately and efficiently. Additionally, the structured use of Postman for CRUD testing illustrated industry-standard testing practices, ensuring the system is robust enough for deployment in manufacturing process optimization contexts.

These results highlight the successful application of theoretical knowledge from the *Pengujian Unit dan Modul Proyek* syllabus into a practical, industry-aligned setting strengthening backend reliability, data accuracy, and system maintainability.

CHAPTER VI

MBKM - Softskill: Kemampuan Bekerjasama dan Kolaborasi

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Course	: Kemampuan Bekerjasama dan Kolaborasi
Course Code	: MII21-4012
Total Credits	: 2 SKS
Internship Title	: MBKM Mandiri OPPO Indonesia
Internship Duration	: January 2025 – April 2025

6.1.Syllabus

Teamwork and Collaboration Skills refer to the motivation and ability to work together with others, as well as the willingness to be part of a group in accomplishing tasks.

1. Solve problems efficiently while collaborating.
2. Actively listen to ideas, suggestions, and feedback from colleagues.
3. Be open to ideas from others or teammates.
4. Maintain effective communication with colleagues.
5. Appreciate others' achievements, encourage them, and make them feel valued.

6.2.Activities

During the internship at OPPO Indonesia, a wide range of teamwork and collaboration activities were undertaken as part of the development and management of both the 360 Peer-to-Peer Grading System and PQPR Software Project. These collaborative activities were integral to the smooth execution of project workflows:

1. **Collaborative Software Development via GitHub**
   All development tasks were coordinated using GitHub repositories and GitHub Projects. Team members worked together on shared branches, created pull requests, conducted code reviews, and resolved merge conflicts, all of which required strong collaboration and mutual support.

2. **Joint Problem-Solving and Debugging Sessions**
   When encountering complex bugs or integration issues, team members worked together to analyze problems, propose solutions, and implement fixes. These sessions promoted real-time knowledge sharing and collective troubleshooting.

3. **Task Sharing and Resource Management**
   Project tasks were divided based on expertise and workload balance. For example, some members handled backend development (APIs, databases), while others focused on frontend or documentation. Coordination ensured complementary work without overlap.

6.3.Fundamental Concepts Related to The Syllabus

The collaborative activities aligned directly with the core concepts of the *Kemampuan Bekerjasama dan Kolaborasi* syllabus:

1. **Problem Solving in Teams**
   Complex development challenges, such as database schema adjustments or API integration issues, were solved through joint brainstorming and task coordination. Collaborative problem-solving ensured efficient issue resolution.

2. **Active Listening and Constructive Feedback**
   During discussions, team members actively listened to suggestions and concerns, acknowledged alternative viewpoints, and provided constructive feedback. This dynamic facilitated mutual understanding and well-informed decisions.

3. **Openness to Ideas**
   An open and inclusive mindset was maintained throughout the internship. All team members were encouraged to contribute ideas freely, whether for improving system architecture, optimizing workflows, or enhancing user experience.

4. **Effective Communication in Teams**
   Regular stand-ups, GitHub discussions, and informal chats ensured continuous, clear communication. This minimized task duplication, clarified expectations, and promoted alignment toward project milestones.

6.4.Result and Discussion

The collaborative practices during the internship generated tangible benefits and strengthened team performance:

1. **Increased Development Efficiency**
   Task distribution, joint problem-solving, and seamless communication ensured efficient progress. Features were delivered on schedule, and system integration proceeded smoothly without major conflicts.

2. **Enhanced Knowledge Sharing and Skill Development**
   Collaborative discussions and peer support accelerated learning. Team members expanded their understanding of tools like FastAPI, Next.js, and PostgreSQL through shared insights and experiences.

3. **Stronger Team Cohesion and Morale**
   Mutual support, active engagement, and encouragement fostered a positive team

culture. Team members felt valued and motivated, leading to sustained collaboration throughout the internship duration.

4. **Better Quality Deliverables**

   Collaborative code reviews and cross-functional coordination contributed to higher quality outputs, with fewer bugs, clearer documentation, and robust system functionality.

5. **Preparation for Professional Teamwork**

   The experience provided realistic exposure to industry-level teamwork, preparing participants for future collaboration in professional IT environments.

6.5. Conclusion

The teamwork and collaboration activities conducted during the MBKM Mandiri internship at OPPO Indonesia effectively strengthened both technical project execution and interpersonal competencies. By practicing structured task sharing, joint problem-solving, open communication, and mutual encouragement, the internship team successfully delivered complex backend systems in a cohesive and efficient manner.

The application of collaborative principles from the *Kemampuan Bekerjasama dan Kolaborasi* syllabus not only contributed to project success but also fostered a supportive and productive team environment. These collaborative experiences have equipped the participant with essential teamwork skills that are crucial for continued success in the software development industry and beyond.

CHAPTER VII

MBKM - Softskill: Kemampuan Berkomunikasi

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MBKM Program Supervisor	: Dr. Azhari, M.T

Course	: Kemampuan Berkomunikasi
Course Code	: MII21-4010
Total Credits	: 2 SKS

Internship Title	: MBKM Mandiri OPPO Indonesia
Internship Duration	: January 2025 – April 2025

7.1.Syllabus

Communication skills encompass a set of activities that ultimately result in high-quality public performance. Effective communication helps overcome diversity, build trust and respect, and create conditions for sharing creative ideas and solving problems.

1. Develop emotional intelligence, including the ability to understand and manage emotions, enabling effective communication, stress management, overcoming challenges, and demonstrating empathy toward others.
2. Communicate clearly and concisely.
3. Possess confidence, empathy, respect, and an open-minded attitude.
4. Develop active listening skills to engage effectively in conversations.
5. Ask thoughtful and meaningful questions.

7.2.Activities

Throughout the internship at OPPO Indonesia, various activities were conducted to strengthen communication skills in both formal and informal professional settings. These activities took place during the execution of the 360 Peer-to-Peer Grading System and PQPR Software Project and were embedded across different stages of project collaboration:

1. **Team Meetings and Stand-Ups**
   Weekly online and offline team meetings were held to discuss project updates, synchronize tasks, and resolve issues. These sessions required clear and concise verbal communication, status reporting, and active participation in discussions.

2. **Issue Discussions and Documentation on GitHub**
   Communication occurred extensively through written mediums via GitHub Issues, Pull Request comments, and project documentation. Articulating technical problems, proposing solutions, and clarifying requirements through written communication was a key part of daily workflows.

3. **Code Review and Feedback Exchange**
   During peer code reviews, feedback was provided to teammates constructively and respectfully. This process involved evaluating code quality, explaining suggestions clearly, and engaging in open dialogue to refine solutions.

7.3.Fundamental Concepts Related to The Syllabus

The communication activities during the internship applied several key concepts from the *Kemampuan Berkomunikasi* syllabus:

1. **Emotional Intelligence and Empathy**
   Active engagement in team meetings and collaborative tasks required understanding teammates’ perspectives, being patient during problem-solving discussions, and managing emotions constructively under deadlines or pressure.

2. **Clear and Concise Communication**
   Throughout GitHub issues, project documentation, and presentations, the ability to express ideas succinctly and logically was critical. This ensured that tasks, problems, and decisions were easily understood and actionable.

3. **Confidence and Open-Mindedness**
   Presenting project updates to supervisors and peers demanded confidence, while receiving feedback with an open-minded attitude promoted productive dialogue and continuous improvement.

4. **Active Listening**
   During meetings and code reviews, attentively listening to teammates' inputs, questions, or concerns fostered mutual understanding, reduced miscommunication, and built trust within the team.

5. **Asking Thoughtful and Meaningful Questions**
   Clarifying task requirements, understanding project expectations, and identifying blockers were accomplished by asking precise and purposeful questions during discussions and reviews.

7.4. Result and Discussion

The consistent practice of communication skills throughout the internship led to several impactful outcomes:

1. **Improved Collaboration and Productivity**
   Clear task descriptions, constructive feedback, and well-documented discussions ensured alignment within the team. Tasks progressed smoothly with minimal confusion, enabling efficient project execution.

2. **Enhanced Team Cohesion**
   Active listening, empathy, and open-minded discussions strengthened trust and

rapport among team members, fostering a collaborative and positive working environment.

3. **Effective Stakeholder Engagement**
   Regular updates and clear presentations to supervisors ensured that stakeholders were well-informed about project status, timelines, and challenges. This proactive communication built credibility and demonstrated accountability.

4. **Professional Growth in Communication Competence**
   The experience enhanced both verbal and written communication skills, improved confidence in expressing technical ideas, and refined the ability to adapt messaging to different audiences (technical peers, supervisors, or non-technical collaborators).

7.5.Conclusion

The communication skills developed and applied during the MBKM Mandiri internship at OPPO Indonesia significantly contributed to both personal growth and project success. By practicing clear, empathetic, and structured communication in meetings, documentation, feedback exchanges, and presentations, collaboration efficiency and team cohesion were greatly enhanced.

The integration of emotional intelligence, active listening, thoughtful questioning, and confidence in expression not only aligned with the *Kemampuan Berkomunikasi* syllabus but also prepared the participant for effective professional communication in real-world IT industry settings. These skills will continue to serve as a vital foundation for future career development and collaborative endeavors in multidisciplinary environments.

CHAPTER VIII
CONCLUSION AND SUGGESTION

The MBKM Mandiri internship program at OPPO Indonesia, conducted from January to April 2025, has been an invaluable opportunity to apply academic knowledge to real-world industrial scenarios. Through the implementation of two major projects the 360 Peer-to-Peer Grading System and the PQPR Software Project this internship fostered both technical proficiency and professional growth.

The projects provided exposure to various facets of backend development, including system design, database management, secure authentication, and performance optimization. Working collaboratively in a dynamic corporate environment enabled the development of soft skills such as project management, teamwork, and effective communication. The collaborative workspace served as the foundation for continuous innovation and knowledge exchange among team members.

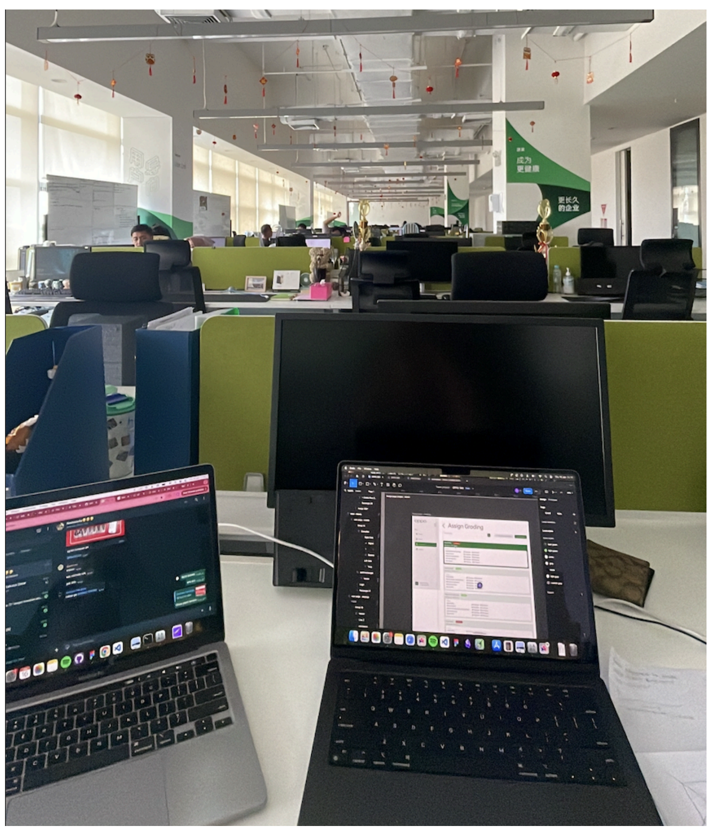

Figure 9.1 Personal Workspace

Throughout the program, clear project planning and execution were demonstrated through systematic task management on GitHub Projects, effective communication strategies, and collaborative problem-solving. The successful presentation of project deliverables to the board of directors, division members, and other stakeholders reflected not only the technical accomplishments but also the capability to communicate complex ideas concisely and confidently.

Figure 9.2 Project Presentation to Board Members & Directors

Beyond technical work, the internship cultivated meaningful professional relationships and cultural understanding within the organization. The closing farewell dinner symbolized the culmination of a rewarding and enriching experience, emphasizing the spirit of camaraderie and appreciation that defined the internship journey.

Figure 9.3 Farewell Dinner Last Group Photo

In summary, the program strengthened both hard and soft skills, bridging the gap between academic learning and industrial application, and preparing the participant for future roles in the technology sector.

Based on the experiences and observations during the internship, the following suggestions are offered to enhance the MBKM Mandiri program and future internship engagements at OPPO Indonesia:

1. **Structured Mentorship Program**
   Implementing a formal mentorship structure where interns are paired with experienced mentors throughout the program would facilitate continuous guidance, skill development, and career advice, further enriching the learning experience.
2. **Continued Alumni Engagement**
   Creating an alumni network or periodic gathering for past interns may foster long-term professional connections and encourage knowledge sharing across cohorts.

By integrating these suggestions, OPPO Indonesia and similar organizations can further enhance the value of their internship programs, supporting the development of skilled, adaptable, and industry-ready graduates.

CHAPTER IX
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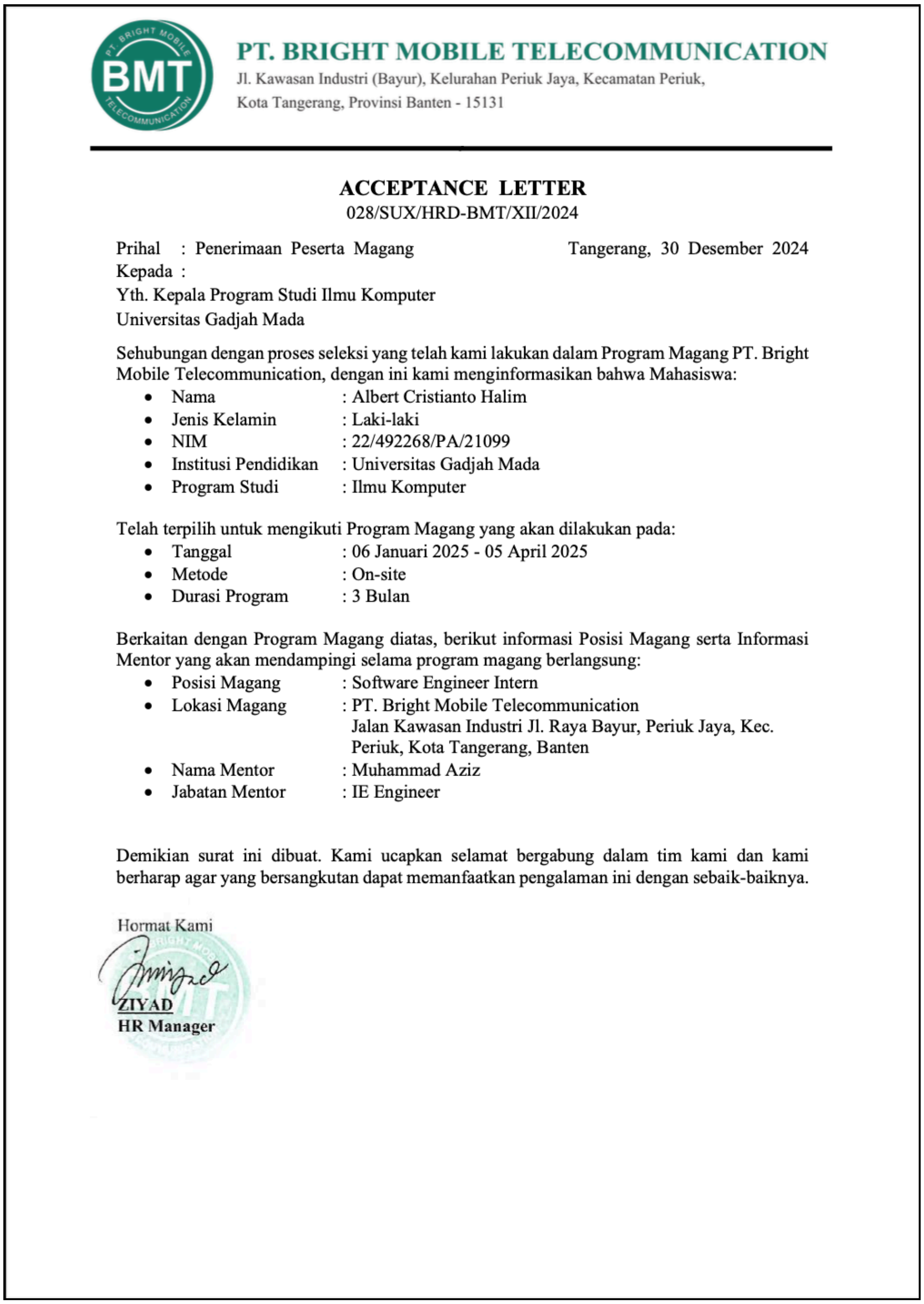
CHAPTER X
APPENDIX

9.1.Letter of Acceptance (LOA)



PT. BRIGHT MOBILE TELECOMMUNICATION
Jl. Kawasan Industri (Bayur), Kelurahan Periuk Jaya, Kecamatan Periuk,
Kota Tangerang, Provinsi Banten - 15131

ACCEPTANCE LETTER
028/SUX/HRD-BMT/XII/2024

Prihal : Penerimaan Peserta Magang — Tangerang, 30 Desember 2024
Kepada :
Yth. Kepala Program Studi Ilmu Komputer
Universitas Gadjah Mada

Sehubungan dengan proses seleksi yang telah kami lakukan dalam Program Magang PT. Bright Mobile Telecommunication, dengan ini kami menginformasikan bahwa Mahasiswa:
- Nama : Albert Cristianto Halim
- Jenis Kelamin : Laki-laki
- NIM : 22/492268/PA/21099
- Institusi Pendidikan : Universitas Gadjah Mada
- Program Studi : Ilmu Komputer

Telah terpilih untuk mengikuti Program Magang yang akan dilakukan pada:
- Tanggal : 06 Januari 2025 - 05 April 2025
- Metode : On-site
- Durasi Program : 3 Bulan

Berkaitan dengan Program Magang diatas, berikut informasi Posisi Magang serta Informasi Mentor yang akan mendampingi selama program magang berlangsung:
- Posisi Magang : Software Engineer Intern
- Lokasi Magang : PT. Bright Mobile Telecommunication
  Jalan Kawasan Industri Jl. Raya Bayur, Periuk Jaya, Kec. Periuk, Kota Tangerang, Banten
- Nama Mentor : Muhammad Aziz
- Jabatan Mentor : IE Engineer

Demikian surat ini dibuat. Kami ucapkan selamat bergabung dalam tim kami dan kami berharap agar yang bersangkutan dapat memanfaatkan pengalaman ini dengan sebaik-baiknya.

Hormat Kami

ZIYAD
HR Manager

Document 9.1.1. Letter of Acceptance from Company

9.2.Letter of Recommendation

UNIVERSITAS GADJAH MADA
FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM
Sekip Utara BLS 21 Yogyakarta 55281 Telp. (0274) 513339 Fax. (0274) 513339
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Nomor : 790/UN1/FMIPA.1.1/AKD/PK.01.06/2025 15 Januari 2025
Hal : Permohonan izin praktik kerja lapangan

Yth. Muhammad Aziz
OPPO Indonesia (PT Bright Mobile Telecommunication)

Dengan hormat,

Sehubungan dengan kegiatan Praktik Kerja Lapangan (PKL) mahasiswa Fakultas Matematika dan Ilmu Pengetahuan Alam, Universitas Gadjah Mada di bawah ini

nama	: Albert Cristianto Halim
nomor mahasiswa	: 22/492268/PA/21099
program studi	: S1 Ilmu Komputer
alamat email	: albertcristiantohalim@mail.ugm.ac.id
Nomor Handphone	: 6281290397055

dengan ini memohon kesediaan Bapak/Ibu memberikan izin kepada mahasiswa tersebut agar dapat melaksanakan praktik kerja lapangan pada:

tempat	: Software Engineer Intern.
alamat	: Jl. Rayan Bayur , Tanggerang, Periuk Jaya,Banten.
waktu	: 06 Januari 2025 s.d. 06 April 2025

Demikian permohonan ini kami sampaikan. Kami berharap Bapak/Ibu dapat memberikan izin dan dukungan untuk praktik kerja lapangan tersebut.

Atas perhatian dan kerja sama Bapak/Ibu disampaikan terima kasih.

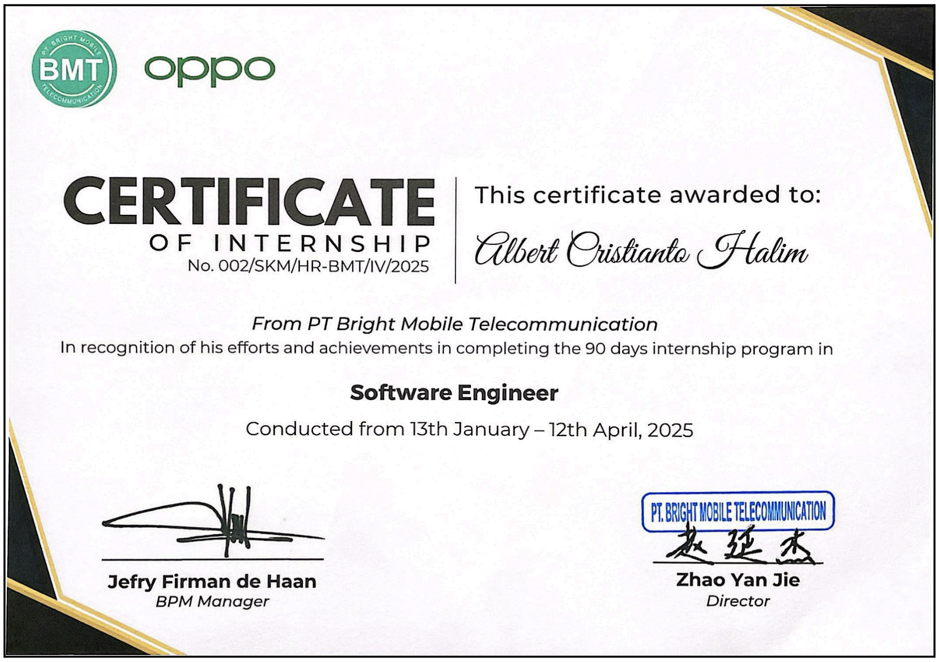
a.n. Dekan,
Wakil Dekan Bidang Pendidikan, Pengajaran dan Kemahasiswaan

Prof. Drs. Roto, M.Eng, Ph.D.
NIP. 196711171993031020

Tembusan :
1. Ketua Program Studi Ilmu Komputer, FMIPA
2. Yang bersangkutan

Document 9.1.2. Letter of Recommendation

9.3.Certificate of Completion



BMT oppo

CERTIFICATE
OF INTERNSHIP
No. 002/SKM/HR-BMT/IV/2025

This certificate awarded to:

Albert Cristianto Halim

From PT Bright Mobile Telecommunication
In recognition of his efforts and achievements in completing the 90 days internship program in

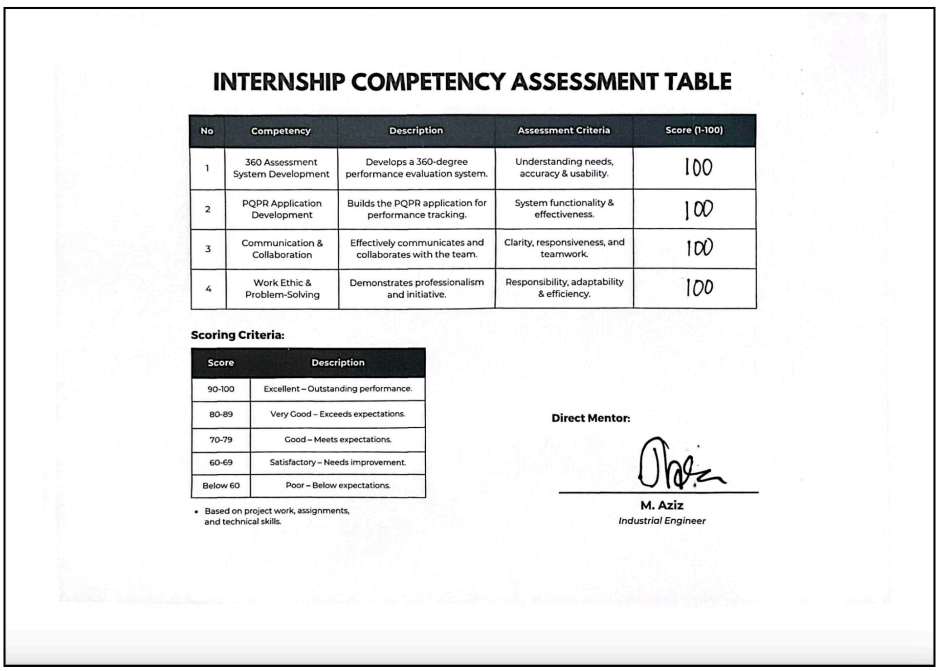
Software Engineer

Conducted from 13th January – 12th April, 2025

PT. BRIGHT MOBILE TELECOMMUNICATION

Jefry Firman de Haan
BPM Manager

Zhao Yan Jie
Director



INTERNSHIP COMPETENCY ASSESSMENT TABLE

No	Competency	Description	Assessment Criteria	Score (1-100)
1	360 Assessment System Development	Develops a 360-degree performance evaluation system.	Understanding needs, accuracy & usability.	100
2	PQPR Application Development	Builds the PQPR application for performance tracking.	System functionality & effectiveness.	100
3	Communication & Collaboration	Effectively communicates and collaborates with the team.	Clarity, responsiveness, and teamwork.	100
4	Work Ethic & Problem-Solving	Demonstrates professionalism and initiative.	Responsibility, adaptability & efficiency.	100

Scoring Criteria:

Score	Description
90-100	Excellent – Outstanding performance.
80-89	Very Good – Exceeds expectations.
70-79	Good – Meets expectations.
60-69	Satisfactory – Needs improvement.
Below 60	Poor – Below expectations.

- Based on project work, assignments, and technical skills.

Direct Mentor:

M. Aziz
Industrial Engineer

Document 9.1.3. Certificate of Completion